Minimal zkVM for Lean Ethereum (draft 0.5.0)

1 What is the goal of this zkVM?

Replacing the BLS signature scheme with a Post-Quantum alternative. One approach is to use stateful hash-based signatures, XMSS, as explained in [1], [2] and [3], and to use a hash-based SNARK to handle aggregation. A candidate hash function is Poseidon2 [4].

We want to be able to:

- **Aggregate** XMSS signatures
- **Merge** those aggregate signatures

The latter involves recursively verifying a SNARK. Both tasks mainly require to prove a lot of hashes. A minimal zkVM (inspired by Cairo [5]) is useful as glue to handle all the logic.

Aggregate / Merge can be unified in a single program, which is the only one the zkVM has to prove (see 1 for a visual interpretation):

Algorithm 1 AggregateMerge

Public input: **pub_keys** (of size n), **bitfield** (k ones, $n-k$ zeros), **msg** (the encoding of the signed message)

Private input: $s > 0$, **sub_bitfields** (of size s), **aggregate_proofs** (of size $s-1$), **signatures**

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                                                    ▷ Bitfield consistency
Check: bitfield = ⋃_{i=0}^{s-1} sub_bitfields[i]
                                    ▷ Verify the first s − 1 sub_bitfields using aggregate_proofs:
for i ← 0 to s − 2 do
    inner_public_input ← (pub_keys, sub_bitfields[i], msg)
    snark_verify("AggregateMerge", inner_public_input, aggregate_proofs[i])
end for
                                          ▷ Verify the last sub_bitfields using signatures
k ← 0
for i ← 0 to n − 1 do
    if sub_bitfields[s-1][i] = 1 then
        signature_verify(msg, pub_keys[i], signatures[k])
        k ← k + 1
    end if
end for
```

2 VM specification

2.1 Field

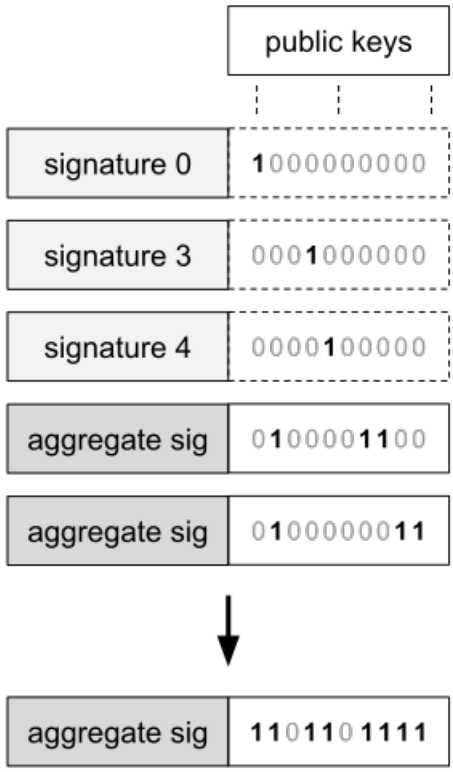
2.1.1 Base field

$\boxed{\text{KoalaBear prime: } p = 2^{31} - 2^{24} + 1}$

Advantages:

- small field → less Poseidon rounds

Figure 1: AggregateMerge visualized.



- $x \to x^3$ is an automorphism of $\mathbb{F}_p^*$, meaning efficient S-box for Poseidon2 (in BabyBear, it's degree 7)
- $< 2^{31} \to$ the sum of 2 field elements can be stored in an u32

The small 2-addicity (24) is not a problem in WHIR, thanks to the use of an interleaved Reed Solomon code (and by playing with the folding factors).

2.1.2 Extension field

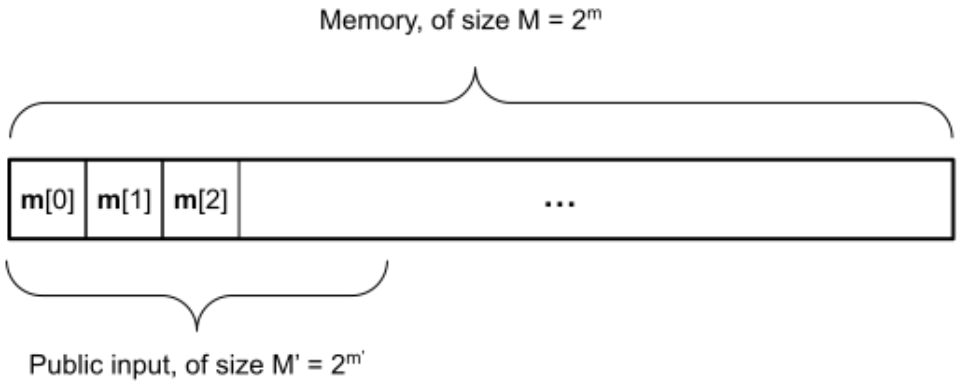
Extension of dimension 5 or 6

Dimension 5 would require the conjecture 4.12 "up to capacity" in WHIR [6] to get ≈ 128 bits of security.

2.2 Memory

- Read-Only Memory
- Word = KoalaBear field element
- Size: $M = 2^m$ with $16 \leq m \leq 29$. The memory size (which depends on the execution) is communicated at the beginning of the proof.
- The first $M' = 2^{m'}$ memory cells hold the "public input," which is written by the verifier and stores the arguments that the leanISA program may receive as input (in our case: message to sign and XMSS public keys), c.f. Figure 2.

Figure 2: Memory structure



2.3 Registers

- $\boxed{\text{pc: program counter}}$
- $\boxed{\text{fp: frame pointer}}$: points to the start of the current stack

Difference with Cairo: no "ap" register (allocation pointer).

2.4 Instruction Set Architecture

Notations:

- α, β and γ represent parameters of the instructions (immediate value operands)
- $\mathbf{m}[i]$ represents the value of the memory at index $i \in \mathbb{F}_p$, with $i < M$ (M: memory size). Any out-of-bound memory access ($i \geq M$) is invalid.
- $\begin{cases} A \\ B \end{cases}$ When using the instruction, either A or B can be used, but not both simultaneously.

2.4.1 ADD / MUL

$a + c = b$ or $a \cdot c = b$ with:

$$a = \begin{cases} \alpha \\ \mathbf{m}[\text{fp} + \alpha] \end{cases} \qquad b = \begin{cases} \beta \\ \mathbf{m}[\text{fp} + \beta] \end{cases} \qquad c = \begin{cases} \text{fp} \\ \mathbf{m}[\text{fp} + \gamma] \end{cases}$$

2.4.2 DEREF

$$\mathbf{m}[\mathbf{m}[\text{fp} + \alpha] + \beta] = \begin{cases} \gamma \\ \mathbf{m}[\text{fp} + \gamma] \\ \text{fp} \end{cases}$$

2.4.3 JUMP (Conditional)

$$\text{condition} = \begin{cases} \alpha \\ \mathbf{m}[\text{fp} + \alpha] \end{cases} \in \{0, 1\} \qquad \text{dest} = \begin{cases} \beta \\ \mathbf{m}[\text{fp} + \beta] \end{cases} \qquad \text{next(fp)} = \begin{cases} \text{fp} \\ \mathbf{m}[\text{fp} + \gamma] \end{cases}$$

$$\text{next(pc)} = \begin{cases} \text{dest} & \text{if condition} = 1 \\ \text{pc} + 1 & \text{if condition} = 0 \end{cases}$$

2.4.4 2 Precompiles

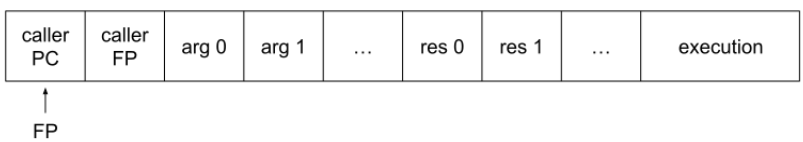
1. **POSEIDON**: Poseidon2 permutation over 16 field elements.
2. **DOT_PRODUCT**: Computes a dot product between either:
   - two slices of extension field elements
   - one slice of base field elements, and one slice of extension field elements

2.5 ISA programming

2.5.1 Functions

1. Each function has a deterministic memory footprint: the length of the continuous frame in memory that is allocated for each of the its calls.
2. At runtime, each time we call our function, we receive via a memory cell a hint pointing to a free range of memory. We then store the current values of pc / fp at the start of this newly allocated frame, alongside the function's arguments, we can then jump, to enter the function bytecode, and modify fp with the hinted value. The intuition here is that the verifier does not care where the new memory frame will be placed (we use a read-only memory, so we cannot overwrite previous frames). In practice, the prover that runs the program would need to keep the value of the allocation pointer "ap", in order to adequately allocate new memory frames, but there is no need to keep track of it from the versifier's perspective.

Figure 3: Memory layout of a function call



2.5.2 Loops

We suggest to unroll loops when the number of iterations is low, and known at compile time. The remaining loops are transformed into recursive functions (by the leanISA compiler).

2.5.3 Range checks

It's possible to check that a given memory cell is smaller than some value t (for $t \leq 2^{16}$) in 3 cycles.

We denote by $\mathbf{m}[\text{fp} + x]$ the memory cell for which we want to ensure $\mathbf{m}[\text{fp} + x] < t$. We also denote by $\mathbf{m}[\text{fp} + i]$, $\mathbf{m}[\text{fp} + j]$ and $\mathbf{m}[\text{fp} + k]$ 3 auxiliary memory cells (that have not been used yet).

1. $\mathbf{m}[\mathbf{m}[\text{fp} + x]] = \mathbf{m}[\text{fp} + i]$ (using DEREF, this ensures $\mathbf{m}[\text{fp} + x] < M$, the memory size)
2. $\mathbf{m}[\mathbf{m}[\text{fp} + x]] + \mathbf{m}[\text{fp} + j] = (\text{t-1})$ (using ADD)
3. $\mathbf{m}[\mathbf{m}[\text{fp} + j]] = \mathbf{m}[\text{fp} + k]$ (using DEREF, this ensures $t - 1 - \mathbf{m}[\text{fp} + x] < M$)

Given $t \leq 2^{16} \leq M$, $\mathbf{m}[\text{fp} + x] < M$, $t - 1 - \mathbf{m}[\text{fp} + x] < M$, and $M \leq 2^{29} < p/2$, we have: $\mathbf{m}[\text{fp} + x] < t$.

Note: From the point of view of the prover running the program, some hints are necessary (filling the values of $\mathbf{m}[\text{fp} + i]$ and $\mathbf{m}[\text{fp} + k]$ must be done at end of execution).

This idea was pointed out by D. Khovratovich, and is an unplanned use of the DEREF instruction.

Example 2.1. Let's say we want to write a function with 2 arguments $x = \mathbf{m}[\text{fp} + 2]$ and $y = \mathbf{m}[\text{fp} + 3]$ ($\mathbf{m}[\text{fp} + 0]$ and $\mathbf{m}[\text{fp} + 1]$ are used, by convention, to store the caller's pc and fp, to return to the previous context at the end of the function), which perform the following:

1. assert(x < 10)
2. z := x*y + 100
3. assert(z < 1000)

Which can be compiled to:

1. $\mathbf{m}[\text{fp} + 4] = \mathbf{m}[[\text{fp} + 2]]$ // check that x is "small"
2. $\mathbf{m}[\text{fp} + 2] + \mathbf{m}[\text{fp} + 5] = 9$ // compute $9 - x$
3. $\mathbf{m}[\text{fp} + 6] = \mathbf{m}[[\text{fp} + 5]]$ // check that $9 - x$ is "small"
4. $\mathbf{m}[\text{fp} + 7] = \mathbf{m}[\text{fp} + 2] * \mathbf{m}[\text{fp} + 3]$ // compute $x.y$
5. $\mathbf{m}[\text{fp} + 8] = \mathbf{m}[\text{fp} + 7] + 100$ // compute $z = x.y + 100$
6. $\mathbf{m}[\text{fp} + 9] = \mathbf{m}[[\text{fp} + 8]]$ // check that z is "small"
7. $\mathbf{m}[\text{fp} + 8] + \mathbf{m}[\text{fp} + 10] = 999$ // compute $999 - z$
8. $\mathbf{m}[\text{fp} + 11] = \mathbf{m}[[\text{fp} + 10]]$ // check that $999 - z$ is "small"
9. JUMP with next(pc) = $\mathbf{m}[\text{fp} + 0]$, next(fp) = $\mathbf{m}[\text{fp} + 1]$, condition = 1 // return

2.5.4 Switch statements

Suppose we want a different logic depending on the value x of a given memory cell, where x is known to be $< k$ (if the value x comes from a "hint", don't forget to range-check it).

Each of the k different value leads to a different branch at runtime, represented by a block of code. We want to jump to the correct block of code depending on x. One efficient implementation consists in placing our blocks of code at regular intervals, and to jump to a $a + b.x$, where a is the offset of the first block of code (in case $x = 0$), and b is the distance between two consecutive blocks.

Example: During XMSS verification, for each of the v chains, we need to hash a pre-image, a number of times depending on the encoding, but known to be $< w$. Here $k = w$, and the $i - th$ block of code we could jump to will execute i times the hash function (unrolled loop).

3 Proving system

3.1 Execution table

3.1.1 Reduced commitment via logup*

In Cairo each instruction is encoded with 15 boolean flags, and 3 offsets. In the execution trace, this leads to committing to 18 field elements at each instruction.

We can significantly reduce the commitments cost using logup*[7]. In the the execution table, we only need to commit to the pc column, and all the flags / offsets describing the current instruction can be fetched by an indexed lookup argument (for which logup* drastically reduces commitment costs).

3.1.2 Commitment

At each cycle, we commit to 5 (base) field elements:

- pc (program counter)
- fp (frame pointer)
- addr_A, addr_B, addr_C

The following 3 (virtual) columns can be interpreted as indexed lookups, and thus not be committed thanks to logup* (in practice though, the gains are modest compared to the indexed lookup into the bytecode, because memory accesses are less repeated):

- $\text{value}_A = \mathbf{m}[\text{addr}_A]$, $\text{value}_B = \mathbf{m}[\text{addr}_B]$, $\text{value}_C = \mathbf{m}[\text{addr}_C]$

3.1.3 Instruction Encoding

Each instruction is described by 15 field elements:

- 3 operands ($\in \mathbb{F}_p$): operand_A, operand_B, operand_C
- 3 associated flags ($\in \{0,1\}$): flag_A, flag_B, flag_C
- 8 opcode flags ($\in \{0,1\}$): ADD, MUL, DEREF, JUMP, POSEIDON, DOT_PRODUCT
- 2 multi-purpose operands: AUX_1, AUX_2

3.1.4 AIR transition constraints

We use $\boxed{\text{transition constraints of degree 5}}$, but it's always possible to make them quadratic with additional columns in the execution table.

We define the following quantities:

- $\nu_A = \text{flag}_A \cdot \text{operand}_A + (1 - \text{flag}_A) \cdot \text{value}_A$
- $\nu_B = \text{flag}_B \cdot \text{operand}_B + (1 - \text{flag}_B) \cdot \text{value}_B$
- $\nu_C = \text{flag}_C \cdot \text{fp} + (1 - \text{flag}_C) \cdot \text{value}_C$

With the associated constraints: $\forall X \in \{A, B, C\} : (1 - \text{flag}_X) \cdot (\text{address}_X - (\text{fp} + \text{operand}_X)) = 0$

For addition and multiplication:

- $\text{ADD} \cdot (\nu_B - (\nu_A + \nu_C)) = 0$
- $\text{MUL} \cdot (\nu_B - \nu_A \cdot \nu_C) = 0$

When DEREF $= 1$, set $\text{flag}_A = 0$, $\text{flag}_C = 1$ and:

$$\mathbf{m}[\mathbf{m}[\text{fp} + \alpha] + \beta] = \begin{cases} \gamma & \rightarrow \quad \text{AUX} = 1,\ \text{flag}_B = 1 \\ \mathbf{m}[\text{fp} + \gamma] & \rightarrow \quad \text{AUX} = 1,\ \text{flag}_B = 0 \\ \text{fp} & \rightarrow \quad \text{AUX} = 0\ (\text{flag}_B = 1) \end{cases}$$

- $\text{DEREF} \cdot (\text{addr}_C - (\text{value}_A + \text{operand}_C)) = 0$
- $\text{DEREF} \cdot \text{AUX} \cdot (\text{value}_C - \nu_B) = 0$
- $\text{DEREF} \cdot (1 - \text{AUX}) \cdot (\text{value}_C - \text{fp}) = 0$

When there is no jump:

- $(1 - \text{JUMP}) \cdot (\text{next}(\text{pc}) - (\text{pc} + 1)) = 0$
- $(1 - \text{JUMP}) \cdot (\text{next}(\text{fp}) - \text{fp}) = 0$

When JUMP $= 1$, the condition is represented by ν_A:

- $\text{JUMP} \cdot \nu_A \cdot (1 - \nu_A) = 0$
- $\text{JUMP} \cdot \nu_A \cdot (\text{next}(\text{pc}) - \nu_C) = 0$

- $\text{JUMP} \cdot \nu_A \cdot (\text{next(fp)} - \nu_B) = 0$
- $\text{JUMP} \cdot (1 - \nu_A) \cdot (\text{next(pc)} - (\text{pc} + 1)) = 0$
- $\text{JUMP} \cdot (1 - \nu_A) \cdot (\text{next(fp)} - \text{fp}) = 0$

Note: the constraint $\text{JUMP} \cdot \nu_A \cdot (1 - \nu_A) = 0$ could be removed, as long as it's correctly enforced in the bytecode.

3.2 Interactions (buses)

Each precompile has a dedicated flag in the instruction encoding. For instance, the instruction corresponding the following Poseidon permutation over 16 field elements:

$$\mathit{Poseidon}(\underbrace{\mathbf{m}[24], \ldots, \mathbf{m}[31]}_{\mathbf{m}[3\cdot 8..4\cdot 8]}, \underbrace{\mathbf{m}[80], \ldots, \mathbf{m}[87]}_{\mathbf{m}[10\cdot 8..11\cdot 8]}) = \underbrace{(\mathbf{m}[\mathbf{m}[\text{fp} + 17] \cdot 8], \ldots, \mathbf{m}[\mathbf{m}[\text{fp} + 17] \cdot 8 + 15])}_{\mathbf{m}[\mathbf{m}[\text{fp}+17]\cdot 8..(\mathbf{m}[\text{fp}+17]+2)\cdot 8]}$$

... is encoded by:

- POSEIDON = 1
- $\text{flag}_A = 1$, $\text{operand}_A = 3$
- $\text{flag}_B = 1$, $\text{operand}_B = 10$
- $\text{flag}_C = 0$, $\text{operand}_C = 17$

Bus 1: Execution table → precompiles (logup GKR)
Bus 2: Execution table / Precompiles → memory (logup GKR)
Bus 3: Instruction fetching from bytecode → execution table (logup*)

TODO: details

Note: In order to make the proof size even smaller, we could remove logup* (using logup GKR everywhere) at the cost of a slower prover (because we would have to commit to the various instruction columns). TODO Explore

4 Annex A: batched logup*

We use the same notations as in the logup* paper [7], but instead of a single evaluation point r, we now have two of them: r_1 and r_2, and two corresponding claims: $I^*T(r_1) = e_1$ and $I^*T(r_2) = e_2$.

1. Use a random challenge α to reduce both claims to: $I^*T(r_1) + \alpha \cdot I^*T(r_2) = e_1 + \alpha \cdot e_2$
2. Commit to $I_*(eq_{r_1} + \alpha \cdot eq_{r_2})$
3. Notice that $I^*T(r_1) + \alpha \cdot I^*T(r_2) = \langle I^*T, eq_{r_1} + \alpha \cdot eq_{r_2} \rangle = \langle T, I_*(eq_{r_1} + \alpha \cdot eq_{r_2}) \rangle$
4. Run a sumcheck for $\langle T, I_*(eq_{r_1} + \alpha \cdot eq_{r_2}) \rangle = e_1 + \alpha \cdot e_2$
5. Open T and $I_*(eq_{r_1} + \alpha \cdot eq_{r_2})$

   Finally, ensuring that $I_*(eq_{r_1} + \alpha \cdot eq_{r_2})$ was correctly committed can be easily adapted from the original protocol, by simply using $X \leftarrow eq_{r_1} + \alpha \cdot eq_{r_2}$.

5 Annex B: simple packing of multilinear polynomials

Note 1: It's always possible to reduce n claims about a multilinear polynomial to a single one, using sumcheck. But this trick is not necessary with WHIR, which natively supports an arbitrary number of claims about the committed polynomial. Given the fact that logup* also has a similar capacity (see 4), the current proof system never requires this n-to-1 reduction.

Note 2: Crucially, the proving cost to add an equality constraint of the form $P((x_1, \ldots, x_n)) = y$ to a committed polynomial P via WHIR is $O(2^k)$ where $k = |\{i, x_i \notin \{0,1\}\}|$ is the number of "non-boolean variables". As a result, adding "sparse" ((x_i) containing boolean values) equality constraints is essentially optimal.

In order to commit to multiple univariate polynomials with FRI, each polynomial must be FFT-ed + Merkle-commited. Even if it's possible to have some batching at the Merkle tree level (see 'MMCS' in Plonky3), the proof size for multiple, complex AIR tables quickly reach the megabyte scale.

With a multilinear PCS (such as WHIR), we can "concatenate" multiple multilinear polynomials into a single one, and commit to it once (offering significant proof size savings).

More details: Given n multilinear polynomials $P_1, \ldots, P_n$ with $\nu_1, \ldots, \nu_n$ variables respectively, we order them from the **largest to the smallest** and concatenate their respective evaluation (on the boolean hypercube):

$$P = [P_1(\{0,1\}^{\nu_1}) \| P_2(\{0,1\}^{\nu_2}) \| \ldots \| P_n(\{0,1\}^{\nu_n})]$$

After padding with zeros to the next power of two, we can interpret the result as the evaluations (on the boolean hypercube) of a multilinear polynomial, with $\nu = \lceil \log_2(\sum_i 2^{\nu_i}) \rceil$ variables, and commit to it.

To reduce an evaluation claim on an "inner" (smaller) polynomial P_i to a claim on the "outer" (larger) polynomial P, we use **boolean selectors**.

Example: Consider 3 multilinear polynomials P_1, P_2, P_3 with 4, 3, and 2 variables respectively. The concatenated polynomial P, that we commit, has $5 = \lceil \log_2(2^4 + 2^3 + 2^2) \rceil$ variables.

- $P_1(x_1, x_2, x_3, x_4) = P(0, x_1, x_2, x_3, x_4)$
- $P_2(x_1, x_2, x_3) = P(1, 0, x_1, x_2, x_3)$
- $P_3(x_1, x_2) = P(1, 1, 0, x_1, x_2)$

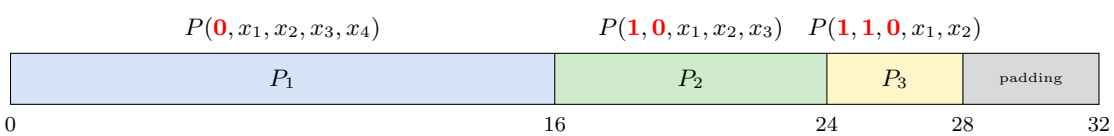


Figure 4: Simple packing of P_1, P_2, P_3 into a single polynomial P

Advantage of this approach: simplicity.

Drawback: padding overhead, i.e. it does not take advantage of the potential repetions at the end of each inner polynomial's evaluations.

There are alterntive ways to handle the packing of multiple multilinear polynomials:

- **Jagged PCS** [8]: No padding overhead, at the cost of an additional sumcheck.
- **Per-polynomial chunking**: decompose the non-repeated par of each inner polynomial into a small (think 3 or 4) number of power-of-two sized chunks, and concatenate all these chunks into a single large polynomial.

Note 3: A meticulous implementation of WHIR could also take advantage of any potential repetitions in the committed polynomial, both at the FFT and the sumcheck level, leaving the Merklelisation as the main overhead.

References

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